

MGTF402: INVESTMENT ANALYSIS

Fall 2026, Master of Quantitative Finance

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COURSE OBJECTIVES

Building on the idea of a risk-return trade-off, this course develops a framework for understanding how investors choose between risky and risk-free financial securities and determine which combinations of assets are desirable. This is the asset allocation or portfolio decision. Finance is inherently a practitioner's science and we emphasize the practical implementation of asset allocation methods: How do we go from the theory to compute actual numbers using a combination of data and subjective judgment? The aim is to provide experience in applying techniques in common use in the portfolio management industry.

A better understanding of portfolio choice requires knowledge of which components of returns can be diversified and which cannot. It also requires understanding how risk premiums are determined, how well they can be estimated and how stable they are through time. To this end we turn to asset pricing models, including the Capital Asset Pricing Model (CAPM) and develop factor pricing models that characterize risk in terms of securities' exposures to a range of common sources of risk, including macroeconomic risk factors. This also leads to the idea of style investing -- accounting for value, growth and size effects -- which is covered in some detail.

How can the techniques of modern portfolio theory help us understand the impact of Environmental, Social, and Governance (ESG) investing? Will pursuing ESG investing reduce portfolio returns or does it, in contrast, hold the promise of better future performance?

Is money left on the table and are our investment models any good in practice? To discuss this, we turn to market efficiency and behavioral finance. We will introduce tools to test market efficiency using so-called event study methodology. If anyone can outperform, presumably it is professional fund managers. We therefore study mutual funds and hedge funds and discuss how to evaluate their performance on a risk-adjusted basis using the asset pricing models developed in the previous part of the course. We discuss recent trends in FinTech towards using AI and algorithms for automated investment decisions.

The course also introduces private markets (private equity, venture capital, and private credit) as an increasingly important allocation for institutional and, increasingly, individual investors, and considers digital assets (cryptocurrencies) as a candidate asset class, evaluating diversification claims against the empirical evidence.

The course next turns to bond pricing, including how to understand the relationship between interest rates at different maturities (the term structure of interest rates), how to measure the risk of a single bond and how to manage the risk of a bond portfolio. How information can be extracted from the term structure of interest rates will also be discussed.

The final topic covered in the course is derivative securities such as options, forwards and futures. We characterize their payoffs, look at their pricing, including binomial lattice methods and the Black-Scholes option pricing formula and discuss how they can be used in investment strategies.

The course engages participants in the empirical analysis of a variety of financial data sets. While most of the assignments can be done in Excel, knowledge of programming languages such as Python is beneficial but not required.

COURSE MATERIALS

Bodie, Z., A. Kane and A.J. Marcus, 2024, Investments, 13th edition. McGraw-Hill Irwin.

We refer to this as BKM. This book provides a comprehensive coverage of a wide range of investment topics and you will be able to use this textbook as a reference for years after you have graduated. The book covers a lot of material and should be considered background reading. Advice on which chapters to concentrate on is provided below.

Preparation of the cases discussed during each weekend session is very important as they have been selected to illustrate the theory we will cover in class. Much of the class discussion will evolve around these cases so active participation is only possible if you have spent a fair amount of time on this material in advance of class. We will also be covering current topics from outlets such as the FT and WSJ.

Students can create accounts and access the Financial Times online subscription here: <https://join.ft.com/dfdd3e48-38b6-4fc0-966e-8c0383cde5b5>

Accounts and access to the Wall Street Journal articles can be created here: <https://ucsd.libguides.com/business/WSJdotcom>

COURSE GRADES

- Assignments (15%)
- Quizzes (35%)
- Final exam (50%)

Grades for the written assignments will partly be based on the effort in answering the questions, partly on the quality of the answer. We encourage you to do the assignments in groups with one report submitted for each group. Please make sure that all names of students in a group are listed on the submitted homework.

The weekly quizzes use Honorlock, an online exam proctoring service that integrates directly with Canvas to help maintain academic integrity in remote and hybrid exams. During a proctored exam, you must use the Chrome browser with the Honorlock extension

TEACHING ASSISTANTS

The teaching assistants for this course are Luka Vulicevic and Max Yang. They will be available to answer questions in person and via e-mail during the week. Their course e-mail is mgtf402.2026@gmail.com.

PROFESSOR CONTACT INFORMATION

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MGTF 402 COURSE OUTLINE

Week 1 — September 28: Understanding risk and returns for asset classes and portfolios

- Course Review
- Measuring risk and returns: Empirical evidence - BKM 5
 - Mean, standard deviation, skewness, kurtosis, drawdown, Value at Risk
 - Digital assets (cryptocurrencies) as a case study in fat tails and extreme drawdowns
- Risk aversion - BKM 6
- Do stocks outperform Treasury Bills? Bessembinder, 2022
- What makes a great stock? Bessembinder, 2024
- Portfolio diversification, asset allocation and construction of efficient portfolios -- BKM 7

Required reading:

BKM chapters 5, 6 and 7.

Background reading:

BKM chapters 2 and 3

Bessembinder, Hendrik, 2022, Do stocks outperform Treasury Bills? Journal of Financial Economics, 2022.

Bessembinder, Hendrik, 2024, Which U.S. stocks generated the highest long-term returns? SSRN working paper #4897069 (Arizona State University).

Week 2 — October 5: Modern Portfolio Theory: Theory and Practice

- Modern Portfolio Theory (continued), BKM 7
- Digital assets: do cryptocurrencies diversify a traditional portfolio?
- Estimation risk, constrained optimization -- Britten-Jones, Jagannathan and Ma, 2003
- ESG investing, ESG frontier -- Pedersen et al., 2020
- Single index model and factor structures -- BKM 8
- Attribution analysis

Assignment 1 due (please see Canvas)

Quiz 1

Required reading:

BKM chapters 7 and 8

Borri, N., Y. Liu, A. Tsyvinski, and X. Wu, 2026, Cryptocurrency as an Investable Asset Class: Coming of Age. Arxiv working paper.

Brinson, G.P., L.R. Hood and G.L. Beebower, 1986, "Determinants of Portfolio Performance," Financial Analysts Journal, 42, 39-44.

Background reading:

Britten-Jones, M., 1999, The Sampling Error in Estimates of Mean-Variance Efficient Portfolio Weights. Journal of Finance 54, 655-671.

Fernandez, P, E. Apellaniz, and J.F. Acin, 2023, Survey: Market risk premium and risk-free rate used for 80 countries in 2023. SSRN discussion paper, April 3, 2023.

HBS case 9-205-040: "Deutsche Bank: Discussing the Equity Risk Premium" (Assignment 1)

Jagannathan, R. and T. Ma, 2003, Risk reduction in large portfolios: why imposing the wrong constraints helps. *Journal of Finance* 58, 1651-1683.

Pedersen, L.H., S. Fitzgibbons, and L. Pomorski, 2021, Responsible investing: The ESG-efficient frontier. *Journal of Financial Economics* 142, 572-597.

Week 3 — October 12: CAPM and Factor Models

- CAPM - BKM 9
- Empirical estimation and tests of CAPM -- BKM 13.1
- Multi-factor models, index models and the APT -- BKM 8, 10
- Investment styles: Value, growth and momentum -- BKM 10.5
- Portfolio sorts
- Economic tracking portfolios -- Lamont, 2001
- Smart Beta: Does factor/smart-beta investing still work? Crowding and post-publication decay (Huang, Song and Xiang, 2024)

Quiz 2

Required reading:

BKM chapter 9, 10, 13.1.

Fama, Eugene F. and French, Kenneth R., 1992, The Cross-Section of Expected Stock Returns, *Journal of Finance* 47, pp. 427-466.

Background reading:

Engle, R.F., S. Giglio, B.T. Kelly, H. Lee, and J. Stroebe, 2020, Hedging Climate Change News. *Review of Financial Studies* 33, 1184-1216.

Huang, S., Y. Song, and H. Xiang, 2024, The Smart Beta Mirage. *Journal of Financial and Quantitative Analysis* 59(6), 2515-2546.

Lamont, O., 2001, Economic tracking portfolios. *Journal of Econometrics* 105, 161-184.

Smith, S. and A. Timmermann (2022), Have Risk Premia Vanished? *Journal of Financial Economics* 145, 553-576.

Week 4 — October 19: Market Efficiency, Behavioral Finance

- Market efficiency -- BKM 11, Christensen et al (2025), Timmermann and Vulicevic (2026)
- Event study methodology -- BKM 11.3
- Behavioral finance -- BKM 12

Assignment 2 due

Quiz 3

Required reading:

BKM chapters 11, 12, 13.1 -- 13.3.

Background reading:

Christensen, K, A. Timmermann, and B. Veliyev, 2025, Warp speed price moves: Jumps after earnings announcements. *Journal of Financial Economics* 16, 1-25.

Timmermann, A. and L. Vulicevic, Compute, Complexity, and the Scaling Laws of Return Predictability. SSRN working paper.

Week 5 — October 26: Performance evaluation and asset management

- Mutual Funds -- BKM 4, 11.5
- Hedge Funds, BKM, 26.1, 26.2
- Portfolio Performance Evaluation -- BKM 24
- Private equity and venture capital: how fund performance is measured (public market equivalent, IRR vs. multiples), the illiquidity premium and the J-curve. Harris, Jenkinson, and Kaplan (2014)

Quiz 4

Required reading:

BKM chapter 4, 11.5, 24 and 26.1, 26.2

Asset managers prepare for investor shift to bespoke equity portfolios. Financial Times, July 27, 2021.

Morningstar unleashes robots to write fund research. Financial Times, March 26, 2021.

Background reading:

Kosowski, R., A. Timmermann, R. Wermers and H. White, 2006, "Can Mutual Funds 'Stars' Really Pick Stocks? New Evidence from a Bootstrap Analysis." Journal of Finance, 61, 2551-2595.

Jones, R.C. and R. Wermers, 2011, "Active Management in Mostly Efficient Markets." Financial Analysts Journal 67, 29-45.

Harris, R.S., T. Jenkinson, and S.N. Kaplan, 2014, Private Equity Performance: What Do We Know? Journal of Finance 69(5), 1851-1882.

Week 6 — November 2: Fixed Income Analysis

- Bond pricing, yields -- BKM 14
- Default Risk -- BKM 14
- Term Structure of Interest rates -- BKM 15
- Forward rates -- BKM 15.2, 15.6
- Risk of Bond portfolios: Duration, convexity -- BKM 16
- Management of Bond portfolios -- BKM 16
- Pension funds -- Asset-liability management
- Bond factor models: Level, slope, and curvature

Assignment 3 due

Quiz 5

Required reading:

BKM, chapters 14-16.

HBS case 9-294-038: "Walt Disney Company's Sleeping Beauty Bonds -- Duration Analysis"

Week 7 — November 9: Fixed Income (cont.). Investment Strategies, Private Markets, and AI in Asset Management

- Dollar cost averaging
- Rebalancing
- Risk parity
- Importance of human capital to investment decisions
- Robo-advisors and machine learning in portfolio construction and advice. D'Acunto, Prabhala, and Rossi (2019), Rossi and Utkus (2020)

Quiz 6

Required reading:

Lecture slides

Background readings:

D'Acunto, F., N. Prabhala, and A.G. Rossi, 2019, The Promises and Pitfalls of Robo-Advising. Review of Financial Studies 32(5), 1983-2020.

Rossi, A.G. and S.P. Utkus, 2020, Who Benefits from Robo-Advising? Evidence from Machine Learning. Working paper.

Week 8 — November 16: Options -- payoffs, trading strategies and basic pricing

- Option payoffs and characteristics -- BKM 20
- European, American options
- Option trading strategies
- Pricing bounds
- Binomial lattice model - Cox, Ross and Rubinstein (1979)

Assignment 4 due

Quiz 7

Required reading:

BKM 20

Background reading:

Cox, John C, Ross, Stephen A., and Rubinstein, Mark, 1979, Option Pricing: A Simplified Approach. Journal of Financial Economics 7, 229-263.

Week 9 — November 23: Option pricing and volatility, futures and forwards

- Black Scholes -- BKM 21
- Understanding Volatility -- BKM 21
- Portfolio Insurance -- BKM 20.3
- Futures and forwards -- BKM 22, 23.2

Quiz 8

Assignment 5 due

Required reading:

BKM 20.3, 21, 22, 23.2

Background reading:

Wall Street Journal, July 13, 2013, Mark Hulbert: Fasten your seat belts: more market turbulence ahead.

Jarrow, R.A., 1999, "In Honor of the Nobel Laureates Robert C. Merton and Myron S. Scholes: A Partial Differential Equation that Changed the World." Journal of Economic Perspectives, 13(4), 229-248.

Week 10 — November 30: Derivatives (cont.): Futures, Forwards and Swaps

- Futures and forwards -- BKM 22, 23.2
- Expectations hypothesis, BKM 22
- Convenience yield, BKM 22

Required reading:

Week 11: Final exam scheduled for Monday, December 7, 8-11am

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. No accommodations can be implemented retroactively.

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.

ACADEMIC INTEGRITY

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<http://senate.ucsd.edu/manual/appendices/appendix2.pdf>